TestPilot — BITS Pilani Final Project Report (Updated 2 Nov 2026)

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TestPilot — BITS Pilani Design Project Report (Evidence Draft)

BITS PILANI | BSDCH ZC229T | TESTPILOT

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

BSc Design and Computing

BSDCH ZC229T · Design Project

TestPilot

AI-Assisted Test Case Generation for a Full-Stack Healthcare Platform

PROJECT REPORT — PRE-SUBMISSION EVIDENCE DRAFT

Reporting date: 31 October 2026

Project mentor: Kavi Arasu V

Academic duration: 8 weeks (fixed)

Team: [Insert verified student names and student IDs]

Submission status: technical implementation and tests verified; required real LLM study, independent participant assessments and final faculty validation remain outstanding. This document deliberately contains no invented experimental observations.

Abstract

TestPilot is a separate web-based tool designed to assist quality assurance engineers in generating, reviewing, and evaluating structured software test cases for an existing open-source healthcare appointment application. The project deploys and tests the reference application rather than recreating it. TestPilot accepts source-grounded patient, doctor and administrative user stories, requests structured cases from an LLM, supports editing and approval decisions, and maps supported approved scenarios to isolated Playwright verification. The system maintains a manual test baseline and records measured regression outcomes, with evidence links and constrained failure screenshots. As of 31 October 2026, the repository contains 25 documented stories spanning 15 use cases, a 36-case manual baseline, and a successful main-branch Playwright run comprising 28 passed tests and one intentionally skipped safeguard test. Six synthetic reviewer-recipe checks also passed. A controlled AI-versus-manual study, ten genuine prompt iterations with reviewer scores, and independent usability sessions must still be completed before final empirical conclusions can be reported.

1. Problem statement and objectives

Manual test design is time-intensive, particularly for role-based systems with authentication, appointment availability, data validation, and state transitions. LLM-generated cases may accelerate initial design but can also miss boundaries, invent unsupported behaviors or provide ambiguous expected results. The project examines AI-assisted test design while retaining human oversight.

Deploy and analyze an existing healthcare application and document at least 15 user stories across patient, doctor and admin roles.

Generate positive, negative and edge/boundary test scenarios using a documented prompt engineering layer.

Allow a QA reviewer to inspect, edit, approve or reject model suggestions before execution.

Run approved, implementable scenarios in a safe isolated testing environment and display measured results.

Compare genuine model-generated and manually designed cases using independent human evaluations and traceable evidence.

2. Project scope and technical implementation

The healthcare application under test is a pre-existing React/Vite frontend and Express/Node.js backend with MongoDB persistence; TestPilot is separate. The deployed components are linked from the evidence register. The practical application frontend uses Bootstrap rather than the proposal’s reference to Redux Toolkit/Tailwind CSS; the separate TestPilot frontend uses HTML/CSS/JavaScript with a Node HTTP server. Those deviations should be acknowledged to faculty as technology selections, not misrepresented as identical to the reference stack.

Module

Implemented role

Boundary

Healthcare AUT

Browsing, registration, booking, doctor/admin feature flows

Existing app, synthetic data in tests

LLM prompt layer

User stories, four audited variants, structured 4–8 cases

Real model access needs private API key/quota

Review workspace

Edit, approve, reject, note gaps, export decisions

Shared-code identity not independent authentication

Playwright runner

Eleven fixed source-reviewed recipes across three stories

No arbitrary LLM code; localhost isolated MongoDB

Reporting

PASS/FAIL, case IDs, CI evidence URLs, bounded failure PNG

Screenshot only when real browser failure captured

3. Study design and reproducibility

The proposed academic comparison uses three purposively selected user stories: US-001 (registration), US-005 (doctor browsing) and US-017 (appointment booking). For each story, the 36-case manual baseline remains frozen, and genuine model suggestions are evaluated against the same source-grounded acceptance criteria. Reviewer decisions distinguish Covered, Not covered and Not reviewed; criterion counts must disclose the number unreviewed. Two independent reviewers should cite candidate IDs and steps and resolve disagreements through documented discussion. Scores for completeness, accuracy and healthcare relevance are each 1–5 and must be entered by actual reviewers, not inferred from automation.

The prompt-engineering study requires ten actual calls covering all three stories. Each retained record must contain story input, prompt variant, reason for the change, model, UTC timestamp, unedited LLM response, measured generation duration, human ratings and reviewer observation. The app supplies the experiment capture interface; no verified completed ten-run dataset was available in the inspected repository at authoring time.

4. Verified technical results (2 November 2026)

Evidence item

Observed result

Source

Documented requirements

25 stories, 15 use cases

testpilot-lab/user-stories.json

Manual comparison baseline

36 test cases

testpilot-lab/manual-baseline.json

Isolated reviewer-recipe verification

6 passed; synthetic, not human-approved

Main-branch CI job logs

Playwright regression suite

29 passed, 1 intentionally skipped, 0 failed

GitHub Actions #41327612679

Frontend and backend checks

Both successful in same main-branch workflow

GitHub Actions #41327612679

Original Gemini case mapping

Four US-017 AI candidate intents mapped to fixed assertions

WEEK7_GEMINI_PLAYWRIGHT_EVIDENCE.md

The six recipe tests are separately invoked in the same CI workflow and should not be added blindly to the 29-test suite figure as an independent count of distinct test cases. Reported PASS statuses refer to actual isolated assertions. The intentionally skipped check exercises the no-approved-plan safety boundary.

5. AI-assisted versus manual test-case evaluation

An internal evidence record dated 26 October 2026 (TestPilot_AI_Comparison_2026-10-26.md) documents 12 genuine Gemini candidate cases across US-001, US-005 and US-017, compared with 13 manual baseline cases. Its provisional assistant-scored coverage was 4 of 6 acceptance criteria for AI and 6 of 6 for manual design. This is a useful exploratory finding, NOT an independently confirmed human evaluation or evidence of ten prompt iterations. The archived US-017 output did not include the later requirement-derived concurrency edge scenario; the sampled AI outputs also contained incomplete doctor-filtering assertions and potentially invalid synthetic registration phone inputs.

User story

Gemini designs

Manual cases

AI preliminary criteria

Manual preliminary criteria

US-001 registration

4

6

2 of 2

2 of 2

US-005 doctor browsing

4

3

0 of 2

2 of 2

US-017 booking

4

4

2 of 2

2 of 2

TOTAL

12

13

4 of 6

6 of 6

These provisional criteria are from an earlier assistant assessment of generated steps and manual scenario titles; actual reviewers must independently validate each judgment, especially cases with weak or confounded test data. No AI superiority, time-saving percentage, or definitive comparative accuracy is claimed.

Metric

Valid reporting rule

Current status

Criterion coverage

Covered / (Covered + Not covered), disclose unreviewed

Not yet faculty-validated

Test design quality

Human 1–5 scores, evidence-based written justification

Awaiting genuine scores

Generation time

Measured provider response duration per actual run

Ten-run study pending

Human review time

Measured independent reviewer durations

Peer study pending

AI candidate execution

Case ID → approved recipe → CI run → PASS/FAIL

Verified for supported mapping only

6. Security, privacy and ethical considerations

No real patient data are required or authorized for test execution; synthetic patients, doctors and appointments are used in an isolated local MongoDB database.

Model-authored scripts, URLs and arbitrary commands are not executed directly. Reviewed case intents can be mapped only to fixed, code-reviewed Playwright recipes.

Generation receipts and review decisions are checked before supported execution; a shared access code does not independently prove reviewer identity.

Failure screenshots are captured only from real failed isolated browser tests, with bounded image size and source-linked artifacts.

Peer feedback is collected with informed participant consent; real participation and sign-off must not be simulated.

7. Ten-iteration prompt engineering plan

The following table is the controlled experimental protocol and NOT a record of ten completed model calls. Each row becomes evidence only when a real response and human scoring record is attached.

Run

Story

Variant

Comparison focus

Evidence

01

US-001

Baseline

Initial registration coverage

PENDING

02

US-001

Boundary

7/8-character password conditions

PENDING

03

US-001

Assertions

Verifiable registration outcomes

PENDING

04

US-005

Baseline

Initial approved directory

PENDING

05

US-005

Boundary

Empty directory and missing records

PENDING

06

US-005

Assertions

Observable directory contract

PENDING

07

US-017

Baseline

Pending booking happy path

PENDING

08

US-017

Boundary

Missing fields / unavailable slot

PENDING

09

US-017

Authorization

Anonymous booking denial

PENDING

10

US-017

Assertions

HTTP 201/409 and persisted state

PENDING

Required per iteration: model/provider, exact story prompt and changed instruction, UTC time, original candidate JSON, number and categories of cases, generation duration, reviewer identity, 1–5 ratings (completeness/accuracy/healthcare relevance), specific rationale, review duration and signed approval/remarks.

8. Independent review and peer usability

Each independent reviewer should inspect the same immutable AI case set and separately rate both AI and manual cases against every acceptance criterion. Disagreements require a documented resolution. At least two consenting peer participants should complete the core workflow and provide measured task completion and usability feedback. The software has forms and validation for those activities; actual participant records remain pending.

Human evidence

Required record

Status

Reviewer A

Case-set hash, per-criterion AI/manual judgment, cited steps, initials

PENDING

Reviewer B

Same source set, independent judgments, initials

PENDING

Disagreement resolution

Each conflict, decision and rationale signed by reviewers

PENDING

Peer session 1

Consent, measured task outcomes, feedback, duration

PENDING

Peer session 2

Consent, measured task outcomes, feedback, duration

PENDING

9. Proposal acceptance and remaining validation

Proposal milestone

Assessment as of 2 November 2026

Existing healthcare deployment, 15+ stories

IMPLEMENTED (25 stories; existing app)

Prompt layer and LLM test candidates

IMPLEMENTED; real 10-iteration study not evidenced

Human case edit/approve/reject

IMPLEMENTED in the Lab

Approved test execution and dashboard

IMPLEMENTED for fixed reviewed scenarios; screenshots when available

Ten scored real experiments

OUTSTANDING: genuine outputs and scores

Independent case evaluation

OUTSTANDING: two participants and resolutions

Peer usability

OUTSTANDING: observed consenting sessions

Faculty-approved final empirical conclusions

OUTSTANDING: team and mentor validation

10. Discussion, limitations and conclusion

The implementation demonstrates the feasibility of combining structured LLM output with a controlled human-approval workflow and Playwright-based verification for a healthcare application. Its strongest completed contribution is source-grounded traceability and safe deterministic execution in an isolated environment, supported by successful CI results. The principal methodological limitations are the restricted set of executable recipes, lack of authenticated reviewer identities, browser-session-based experiment storage, prompt/API quota dependency, and absence of a completed independent human empirical study.

The prototype is technically ready for a controlled demonstration. It is not yet justified to claim that LLM test design is faster, more accurate or more complete than manual design, or that all faculty outcomes are achieved. Completing the ten authentic experiments and participant assessments, and attaching the evidence to this document, will permit final academic evaluation. These steps are human evidence collection tasks rather than software implementation defects.

References and traceable artifacts

BITS Pilani WILP. BSDCH ZC229T Design Project – supplied project proposal (primary acceptance contract).

Copeland, L. (2004). A Practitioner’s Guide to Software Test Design. Artech House.

Playwright documentation: https://playwright.dev/docs/intro

OpenAI Prompt Engineering Guide: https://platform.openai.com/docs/guides/prompt-engineering

TestPilot repository: https://github.com/Akash-annavarapu/doctor-appointment-booking-system

Verified main-branch run: https://github.com/Akash-annavarapu/doctor-appointment-booking-system/actions/runs/41327612679

Faculty protocol: docs/BITS_PROPOSAL_ACCEPTANCE_AND_STUDY_PROTOCOL.md

Peer-study evidence: docs/PEER_REVIEW_SAFE_EXECUTION_SCREENSHOT_HANDOFF.md

Archived booking mapping: docs/WEEK7_GEMINI_PLAYWRIGHT_EVIDENCE.md

Appendix A. Final validation and sign-off checklist

Validation check

Evidence identifier / sign-off

Status

10 real LLM responses across ≥3 user stories

[Attach JSON export, hashes, date]

PENDING

All 10 responses scored by real humans

[Attach score sheet and reviewer notes]

PENDING

Two independent AI/manual evaluations

[Attach comparison exports]

PENDING

Disagreements adjudicated

[Attach rationale and both initials]

PENDING

Two consent-based peer usability sessions

[Attach session records and evidence]

PENDING

CI pass/fail evidence

GitHub Actions #41327612679

VERIFIED

Final mentor acceptance

[Date, mentor signature/approval]

PENDING

Do not replace pending entries with guessed outcomes. Evidence should be independently verifiable and restricted to synthetic healthcare records.

Appendix B. Evidence reconciliation (2 November 2026)

The current published TestPilot Lab deployment is live at https://testpilot-lab.onrender.com/. Render reports commit 042e14ae (guided ten-run experiment and per-iteration review interface) deployed successfully on 1 November 2026. GitHub main-branch workflow 41327612679 completed successfully: 29 Playwright tests passed, zero failed and one intentionally skipped guard. The six separate synthetic reviewed-case recipe checks also passed. These are genuine automated engineering outcomes, not results of the required human experiment.

A historical study record from 26 October 2026 documents 12 real Gemini-generated candidates across the three target user stories (four each). It compares them with 13 manual designs and reports a preliminary AI assessment of 4/6 acceptance criteria covered versus 6/6 in the manual baseline. This remains assistant-reviewed exploratory evidence; it does not count as ten separate prompt iterations and it has not been independently validated by two humans.

At this review, no authenticated export proving all ten fresh Gemini iterations with participant-entered 1–5 ratings was available. No complete two-reviewer per-story exports, no verified consent-based peer observation exports and no signed mentor approval were supplied. Accordingly, the empirical analysis remains pending and the report is a PRE-SUBMISSION DRAFT.

Appendix C. Required closure artifacts

Deliverable

Evidence to attach

Current verified status

Ten real Gemini iterations

Export from Lab study runner; UTC timestamps, prompt variants, original outputs

NOT VERIFIED

Ten human quality ratings

Three 1–5 ratings and written case-specific observation per run

NOT VERIFIED

Independent AI/manual reviews

Two genuine reviewers per story; matching case-set evidence, cited steps, conflict resolutions

NOT VERIFIED

Peer usability

Two consenting participants; observed tasks, measured duration, feedback

NOT VERIFIED

Final faculty approval

Mentor confirmation, date, final report revision

NOT VERIFIED

Technical CI

GitHub workflow 41327612679; 29 Playwright pass, one skip

VERIFIED

Only replace “NOT VERIFIED” after reviewing original provider or participant records. A synthetic browser fixture or AI-authored judgment cannot be submitted as independent human validation. The report is not an assertion of project acceptance by faculty.

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